ECE 310 Fall 2026
Lecture 8: Inverse z-transform

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Learning Objectives

- Write the common forms for the z-transform of finite-length and infinite-length signals.
- Compute the inverse z-transform of a given rational expression in the z-domain using partial fraction expansion and inspection.



Common Forms of z-transform: Finite-length Signals

Any finite-length signal $x[n]$ with start index n_s and end index n_e can be written as a linear combination of scaled and shifted Kronecker delta signals:

$$x[n] = \sum_{k=n_s}^{n_e} x[k]\delta[n-k].$$

$\delta[n] \xleftrightarrow{\mathcal{Z}} 1$

$x[n-k] \xleftrightarrow{\mathcal{Z}} z^{-k}X(z)$

The resulting z-transform of any discrete-time signal is then

$x[k]\delta[n-k] \xleftrightarrow{\mathcal{Z}} x[k]z^{-k}$

$$X(z) = \sum_{k=n_s}^{n_e} x[k]z^{-k}$$



Common Forms of z-transform: Infinite-length Signals

Infinite-length signals have z-transforms given by *rational expressions*, i.e. a fraction with polynomials in the numerator and denominator: $X(z) = B(z)/A(z)$. When $N \leq M$, we have a *proper rational expression*.

Ex: $X(z) = \dfrac{1-2z^{-1}}{1+6z^{-1}+5z^{-2}}$

$$X(z) = \frac{\sum_{k=0}^{N-1} b_k z^{-k}}{1 + \sum_{k=1}^{M} a_k z^{-k}} \tag{1}$$

$X(z) = \dfrac{1-2z^{-1}}{(1+5z^{-1})(1+z^{-1})}$

$$X(z) = \frac{\prod_{k=1}^{N-1}(1 - q_k z^{-1})}{\prod_{k=1}^{M}(1 - p_k z^{-1})} \tag{2}$$

→ zeros: $z = 2$

→ poles: $z = -5$, $z = -1$

q_k = zeros

p_k = poles

Inverse z-transform

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The inverse z-transform for $X(z)$ is given by

$$x[n] = \frac{1}{j2\pi} \oint_C X(z) z^{n-1} dz.$$



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$$x[n] = \frac{1}{j2\pi} \oint_C X(z) z^{n-1} dz.$$

Don't worry, we won't use this formula!

Instead find x[n] by inspection



Computing Inverse z-transform: Finite-length Signals

We have already determined this rule!

$$x[n] = \sum_{k=n_s}^{n_e} x[k]\delta[n-k] \overset{\mathcal{Z}}{\longleftrightarrow} X(z) = \sum_{k=n_s}^{n_e} x[k]z^{-k}$$



Example 1: Inverse z-transform by Inspection

Determine the inverse z-transform of the following z-transforms.

(a) $X(z) = 1 - \frac{1}{4}z^{-1} + \frac{1}{16}z^{-2}$

$$x[n] = \delta[n] - \frac{1}{4}\delta[n-1] + \frac{1}{16}\delta[n-2]$$

$$= \left\{1, -\frac{1}{4}, \frac{1}{16}\right\}$$

(b) $X(z) = \frac{1}{1+\frac{1}{3}z^{-1}}, |z| < \frac{1}{3}$

Left-sided

$a = -\frac{1}{3}$

$\rightarrow x[n] = -\left(-\frac{1}{3}\right)^n u[-n-1]$

Time-shifting: $x[n-k] \longleftrightarrow z^{-k}X(z)$

$\delta[n] \longleftrightarrow 1$

$a^n u[n] \longleftrightarrow \frac{1}{1-az^{-1}}, |z| > |a|$

$-a^n u[-n-1] \longleftrightarrow \frac{1}{1-az^{-1}}, |z| < |a|$

Example 1: Inverse z-transform by Inspection

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Determine the inverse z-transform of the following z-transforms.

(c) $X(z) = \frac{1}{1-\frac{1}{2}z^{-1}} - \frac{3z^{-1}}{1+\frac{1}{5}z^{-1}}, \ |z| > \frac{1}{2}$

$a = \frac{1}{2}$ $\quad a = -\frac{1}{5}$ $\quad$ right-sided

$x[n] = \left(\frac{1}{2}\right)^n u[n] - 3\left(-\frac{1}{5}\right)^{n-1} u[n-1]$

(d) $X(z) = \frac{1-2z^{-1}+3z^{-2}}{(1-z^{-1})(1+\frac{2}{3}z^{-1})(1-2z^{-1})}, \ |z| > 2$

Too difficult (for now)

Time-shifting: $x[n-k] \longleftrightarrow z^{-k}X(z)$

$\delta[n] \longleftrightarrow 1$

$a^n u[n] \longleftrightarrow \frac{1}{1-az^{-1}}, \ |z| > |a|$

$-a^n u[-n-1] \longleftrightarrow \frac{1}{1-az^{-1}}, \ |z| < |a|$

Computing Inverse z-transform: Infinite-length Signals

Consider the following third form of the z-transform for infinite-length signals:

$$X(z) = \frac{\prod_{k=1}^{N-1}(1 - q_k z^{-1})}{\prod_{k=1}^{M}(1 - p_k z^{-1})} = \sum_{k=1}^{M} \frac{A_k}{1 - p_k z^{-1}} \tag{3}$$

$$x[n] = \sum_{k=1}^{M} A_k (p_k)^n u[n]$$ (assuming right-sided)



Computing Inverse z-transform: Partial Fraction Decomposition

We need some procedure for taking forms (1) or (2) of the z-transform in the previous slides and translating them to form (3) in the previous slide. This procedure is known as *partial fraction decomposition*.



Computing Inverse z-transform: Partial Fraction Decomposition

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Partial Fraction Decomposition

Suppose we have z-transform $X(z)$:

1. Factorize the denominator of $X(z)$ to obtain "form (2)" of the z-transform for infinite-length signals and identify each pole of the system. Set this equal to "form (3)" where each pole has it's own rational expression and A_k coefficient.
2. Multiply both sides – $X(z)$ and your new factorized form – by the shared denominator of $X(z)$.
3. Solve for each A_k value by setting $z = p_k$.



Example 2

Determine $x[n]$ corresponding to the following z-transform. You may assume $x[n]$ is right-sided.

$$X(z) = \frac{2 - z^{-1}}{1 + \frac{2}{3}z^{-1} - \frac{1}{3}z^{-2}}$$

① Factorize denominator:

$1 + \frac{2}{3}z^{-1} - \frac{1}{3}z^{-2}$

↑ a ↑ b ↑ c

Roots = poles $= \dfrac{-\frac{2}{3} \pm \sqrt{\frac{4}{9} - 4(1)(-\frac{1}{3})}}{2(1)}$

$= z = -1, \; z = \frac{1}{3}$

$\Rightarrow X(z) = \dfrac{2 - z^{-1}}{(1 + z^{-1})(1 - \frac{1}{3}z^{-1})} = \dfrac{A_1}{1 + z^{-1}} + \dfrac{A_2}{1 - \frac{1}{3}z^{-1}}$

Example 2

Determine $x[n]$ corresponding to the following z-transform. You may assume $x[n]$ is right-sided.

$$X(z) = \frac{2 - z^{-1}}{1 + \frac{2}{3}z^{-1} - \frac{1}{3}z^{-2}}$$

(1) $X(z) = \dfrac{2 - z^{-1}}{(1 + z^{-1})(1 - \frac{1}{3}z^{-1})} = \dfrac{A_1}{1 + z^{-1}} + \dfrac{A_2}{1 - \frac{1}{3}z^{-1}}$

(2) Multiply both sides by denominator: $2 - z^{-1} = A_1\left(1 - \frac{1}{3}z^{-1}\right) + A_2\left(1 + z^{-1}\right)$

(3) A_1: $z = -1$

$2 - (-1)^{-1} = A_1\left(1 - \frac{1}{3}(-1)^{-1}\right) + A_2\left(1 - (-1)^{-1}\right)$

$3 = \frac{4}{3}A_1$

$A_1 = \frac{9}{4}$

A_2: $z = \frac{1}{3}$

$2 - (\frac{1}{3})^{-1} = A_1\left(1 - \frac{1}{3}(\frac{1}{3})^{-1}\right) + A_2\left(1 + (\frac{1}{3})^{-1}\right)$

$-1 = 4A_2$

$A_2 = -\frac{1}{4}$

Example 2

Determine $x[n]$ corresponding to the following z-transform. You may assume $x[n]$ is right-sided.

$$X(z) = \frac{2 - z^{-1}}{1 + \frac{2}{3}z^{-1} - \frac{1}{3}z^{-2}}$$

(1) $X(z) = \dfrac{2 - z^{-1}}{(1 + z^{-1})(1 - \frac{1}{3}z^{-1})} = \dfrac{A_1}{1 + z^{-1}} + \dfrac{A_2}{1 - \frac{1}{3}z^{-1}}$

(2) Multiply both sides by denominator: $2 - z^{-1} = A_1\left(1 - \frac{1}{3}z^{-1}\right) + A_2\left(1 + z^{-1}\right)$

(3) A_1: $z = -1$

$2 - (-1)^{-1} = A_1\left(1 - \frac{1}{3}(-1)^{-1}\right) + A_2\left(1 - (-1)^{-1}\right)$

$3 = \frac{4}{3}A_1$

A_2: $z = \frac{1}{3}$

$2 - \left(\frac{1}{3}\right)^{-1} = A_1\left(1 - \frac{1}{3}\left(\frac{1}{3}\right)^{-1}\right) + A_2\left(1 + \left(\frac{1}{3}\right)^{-1}\right)$

$-1 = 4A_2$

$A_1 = \frac{9}{4}$, $A_2 = -\frac{1}{4}$ $\longrightarrow$ $x[n] = \frac{9}{4}(-1)^n u[n] - \frac{1}{4}\left(\frac{1}{3}\right)^n u[n]$

Example 3

Determine all possible ROCs for the below $X(z)$ and find the corresponding $x[n]$ in each case.

$$X(z) = \frac{1}{\left(1 - \frac{4}{3}z^{-1}\right)\left(1 - \frac{2}{3}z^{-1}\right)}$$



Example 3

Determine all possible ROCs for the below $X(z)$ and find the corresponding $x[n]$ in each case.

Step #1: $X(z) = \dfrac{1}{\left(1-\frac{4}{3}z^{-1}\right)\left(1-\frac{2}{3}z^{-1}\right)} = \dfrac{A_1}{1-\frac{4}{3}z^{-1}} + \dfrac{A_2}{1-\frac{2}{3}z^{-1}}$

Step #2: $1 = A_1\left(1-\frac{2}{3}z^{-1}\right) + A_2\left(1-\frac{4}{3}z^{-1}\right)$

Step #3:

(Pole 1) A_1 $z = \frac{4}{3}$:

$1 = A_1\left(1-\frac{2}{3}\left(\frac{4}{3}\right)^{-1}\right)$

$1 = \frac{1}{2}A_1$

$A_1 = 2$

(Pole 2) A_2 $z = \frac{2}{3}$:

$1 = A_2\left(1-\frac{4}{3}\left(\frac{2}{3}\right)^{-1}\right)$

$A_2 = -1$

Example 3

Determine all possible ROCs for the below $X(z)$ and find the corresponding $x[n]$ in each case.

$$X(z) = \frac{1}{\left(1-\frac{4}{3}z^{-1}\right)\left(1-\frac{2}{3}z^{-1}\right)} = \frac{A_1}{1-\frac{4}{3}z^{-1}} + \frac{A_2}{1-\frac{2}{3}z^{-1}}$$

Step #1:

P_1 \ P_2	RS	LS
RS	$\lvert z\rvert > \frac{4}{3} \cap \lvert z\rvert > \frac{2}{3}$ $\lvert z\rvert > \frac{4}{3}$	$\lvert z\rvert > \frac{4}{3} \cap \lvert z\rvert < \frac{2}{3}$ Empty or $\{\}$ or $\emptyset$
LS	$\lvert z\rvert < \frac{4}{3} \cap \lvert z\rvert > \frac{2}{3}$ $\frac{2}{3} < \lvert z\rvert < \frac{4}{3}$	$\lvert z\rvert < \frac{4}{3} \cap \lvert z\rvert < \frac{2}{3}$ $\lvert z\rvert < \frac{2}{3}$

Step #2: $1 = A_1\left(1-\frac{2}{3}z^{-1}\right) + A_2\left(1-\frac{4}{3}z^{-1}\right)$

Step #3:

(Pole 1)

A_1 $z = \frac{4}{3}$:

$1 = A_1\left(1-\frac{2}{3}\left(\frac{4}{3}\right)^{-1}\right)$

$1 = \frac{1}{2}A_1$

$A_1 = 2$

(Pole 2)

A_2 $z = \frac{2}{3}$:

$1 = A_2\left(1-\frac{4}{3}\left(\frac{2}{3}\right)^{-1}\right)$

$A_2 = -1$

Example 3 (continued)

Determine all possible ROCs for the below $X(z)$ and find the corresponding $x[n]$ in each case.

$$X(z) = \frac{1}{\left(1-\frac{4}{3}z^{-1}\right)\left(1-\frac{2}{3}z^{-1}\right)} = \frac{2}{1-\frac{4}{3}z^{-1}} - \frac{1}{1-\frac{2}{3}z^{-1}}$$

1. (RS, RS): $x_1[n] = 2\left(\frac{4}{3}\right)^n u[n] - \left(\frac{2}{3}\right)^n u[n]$

2. (LS, RS): $x_2[n] = -2\left(\frac{4}{3}\right)^n u[-n-1] - \left(\frac{2}{3}\right)^n u[n]$

3. (LS, LS): $x_3[n] = -2\left(\frac{4}{3}\right)^n u[-n-1] + \left(\frac{2}{3}\right)^n u[-n-1]$

4. (RS, LS): z-transform does not exist for any values of z.
   - $x_4[n]$ would be unbounded as $n \to \infty$ and $n \to -\infty$

Table pairs:

right-sided (RS)

$a^n u[n] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{1-az^{-1}}, \quad |z| > |a|$

$-a^n u[-n-1] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{1-az^{-1}}, \quad |z| < |a|$

left-sided (LS)